

DECODING OF KRYPTOS K4 USING K4 USING C² K4 DOUBLE-MASK-OTP

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ABSTRACT

Using the C2 K4 Double-Mask-OTP Hypothesis, we decoded Jim Sanborn's K4 section of the sculpture Kryptos (1990). Modifications to known cryptography methods were adapted to more tactile analog approaches favorable to an artist, rather than a digital (computational) method used by cryptographers. The hints of BERLIN and CLOCK for positions 64 through 74 were used as a crib in the decoding of K4.

INTRODUCTION

Previously the C2 K4 Double-Mask-OTP Hypothesis for solving K4 for Kryptos was proposed on July 15, 2019. After much trial and error, plaintext was decoded from K4 on July 21, 2019 using one-time pad.

MATERIAL AND METHODS

While the concept is not difficult in theory, the arduous part requires many attempts of manipulating the K4 public key to potentially decode K4. Through reading the dinner notes throughout the years, it was apparent that K4 could be solved with pencil and paper.

In lieu of pencil and paper, Microsoft Excel did the heavy lifting for the transpositions and decoding.

K4 Public Key & BERLINCLOCK Crib

Using the OTP theory, the 2010 NYPVTT decodes to BERLIN¹ immensely supported use of a public key. That is, to decoding the text would require the key MUYKLG. In the 97 character K4 ciphertext there is only a single M and a

single Y present – but also not in the 64th and 66th position respectively but the 70th and 65th position.

In addition, KORNA would also be required to decode MZFPK into CLOCK². Armed with this information a table was created to see if there were any patterns of a transposition and/or substitution.

From the frequency analysis of MUYKLGKORNA:

Position required to decode to plaintext	OTP Decode	Ciphertext position
64	M	70
65	U	5, 11, 55, 88, 90, 92
66	Y	65
67	K	3, 32, 46, 53, 74, 78, 87, 94
68	L	12, 16, 23, 54
69	G	9, 31, 76, 86
70	K	3, 32, 46, 53, 74, 78, 87, 94
71	O	1, 6, 8, 15, 35
72	R	4, 24, 29, 97
73	N	30, 61, 64
74	A	50, 58, 91, 96

In order for some form of transposition to work, public key (K4 ciphertext) Y would have to shift 1 character forward, and M would have to shift back 6 characters. Due to the asymmetrical nature of the shift, it would be extremely difficult for a single columnar transposition, and likely there would be some form of a double transposition.

From Jim Sanborn's prior process, K3 was done by first placing it in a 42 column by 8 row matrices, rotated it by 90 degrees clockwise and then transposed it on a 14 column by 24 row matrices with an additional 90-degree clockwise rotation. The resulting text became the cipher for K3³.

In order to select the row and columns for K3, the values were calculated from the factors of 336 (2, 2, 2, 6, 7) being 2x2x2 and 6x7. For K4, the character count was 97, which would be a prime number. However, including the question mark would increase the character count to 98 with factors of 2, 7, and 7. Since Jim worked with a 31-column grid, it was unlikely he was using a 49x2 matrix, but some variant of 14x7.

¹ <https://www.nytimes.com/2010/11/21/us/21code.html>

² <https://www.nytimes.com/2014/11/21/us/another-kryptos-clue-is-offered-in-a-24-year-old-mystery-at-the-cia.html>

³ <http://www.thekryptosproject.com/kryptos/charts/method.php>

Jefferson Disk Method

It would be plausible to assume that Mr. Sanborn chose a technique that would give him flexibility to manipulate text without much knowledge in mathematics or cryptography. This would be able to explain the shifts between characters for M and Y in the K4 public key as different disks could be reordered to provide a set shift, and each individual letter could be shifted within the group. In our opinion, Mr. Sanborn was very generous to perform his K3 double columnar shift by mirroring the order in lieu of intermittent swaps. Without the help of the BERLIN and CLOCK hints, the order of the disks might not be possible. That is, a one-time pad can be decoded to any plaintext.

In dealing with the question mark for K4, both options were attempted with repeated reordering of columns/disks and decoding. With the question mark included, the K4 ciphertext was rewritten as:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	?	O	B	K	R	U	O	X	O	G	H	U	L	B
2	S	O	L	I	F	B	B	W	F	L	R	V	Q	Q
3	P	R	N	G	K	S	S	O	T	W	T	Q	S	J
4	Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U
5	D	I	A	W	I	N	F	B	N	Y	P	V	T	T
6	M	Z	F	P	K	W	G	D	K	Z	X	T	J	C
7	D	I	G	K	U	H	U	A	U	E	K	C	A	R

Now the Y part of the key is in the correct place, which would decode to P to R. The columns/disks would then be reordered, shifted downwards, and decoded (via OTP) until it would be able to provide some sort of masked plaintext. What is also important is that disk/column 1 contains the M character which needs to be moved to column 8.

This method was then repeated to match the crib of MUYKLG for columns 9-14 and KORNA for columns 1-5. Since each column is grouped into 7 characters, it can be shifted (or turned in a Jefferson disk). Since the crib was given, there would only be a set number of possibilities or orientations available to decode to BERLINCLOCK.

The columns/disks were marked then tabulated for the frequency of letters in MUYKLGKORNA:

Position	OTP Decode	Columns (with the letter)	Column placement required
64	M	1	#9
65	U	5, 6, 7, 9, 12, 14	#10
66	Y	10	#11
67	K	4,4, 5,5,5, 9, 11, 12	#12
68	L	3, 10, 13	#13
69	G	3, 7, 10	#14
70	K	4,4, 5,5,5, 9, 11, 12	#1
71	O	2,2, 7, 8, 9	#2
72	R	2, 5, 11, 14	#3
73	N	3, 6, 9	#4
74	A	3, 8, 9, 13	#5

In order to fit exactly in the correct character position, the M and Y have to fit in position #8 and #10 respectively. What can also be seen through the process of elimination is that L and G can only be columns 3/13 and 3/7 respectively.

Upon trial and error, it was noted that the columns/disks would change a set of characters in fixed patterns. Originally when attempting to decode the ciphertext, we noticed that using a 14-column table would result in one character to change and a group of six would also change. That is, when characters 1-4 are reoriented to columns 11-14 (to match OBKR on K4), the change of the disks would only affect a single column as a whole (plaintext orientation). The pattern was identified as:

JEFFERSON DISK (K4 PUBLIC KEY)	PLAINTEXT 14 COLUMN (8 ROWS) OUTPUT
1	11
2	12
3	13
4	14
5	1
6	2
7	3
8	4
9	5
10	6
11	7
12	8
13	9
14	10

With the possibilities, the combinations were mapped out, reordered, transposed, and decoded (OTP). Since the decoded plaintext was masked, it was very imperative to sound out all the plaintexts. Knowing that the M and Y public key columns were fixed, all the combination of letters were lined up for the 2nd row. If a phrase was made, it was matched to the frequency-column table to ensure there were no conflicts.

Soon enough the sections of BERLINCLOCK starting matching up and we were left with 3 columns (5, 6 and 7). Again, all the character combinations were mapped out and attempts were made to obtain a plausible combination.

To spare everyone reading all possibilities, the order of the columns ended up being:

	2	1	6	8	4	4	3	1	2	0	9	3	7	5
1	O	H	U	X	B	K	L	?	U	G	O	B	O	R
2	O	R	B	W	Q	I	Q	S	V	L	F	L	B	F
3	R	T	S	O	J	G	S	P	Q	W	T	N	S	K
4	S	J	Z	W	U	E	L	Q	K	T	A	S	Z	K
5	I	P	N	B	T	W	T	D	V	Y	N	A	F	I
6	Z	X	W	D	C	P	J	M	T	Z	K	F	G	K
7	I	K	H	A	R	K	A	D	C	E	U	G	U	U

When transposed vertically, the shift is shown as:

	2	1	6	8	4	4	3	1	2	0	9	3	7	5
1					K				G					
2			U		I	L			L					R
3			B		B	G	Q		W					F
4			S		Q	E	S		T		B			K
5		H	Z		J	W	L		U	Y		L		K
6	O	R	N		U	P	T		V	Z		N		I
7	O	T	W	X	T	K	J	?	Q	E	O	S	O	K
	R	J	H	W	C		A	S	K		F	A	B	U
	S	P		O	R			P	V		T	F	S	
	I	X		W				Q	T		A	G	Z	
	Z	K		B				D	C		N		F	
	I			D				M			K		G	
				A				D			U		U	

Letters below row 7 would then just wrap around and return to the table:

	2	1	6	8	4	4	3	1	2	0	9	3	7	5
1	R	J	H	W	C	K	A	S	K	G	F	A	B	U
2	S	P	U	O	R	I	L	P	V	L	T	F	S	R
3	I	X	B	W	B	G	Q	Q	T	W	A	G	Z	F
4	Z	K	S	B	Q	E	S	D	C	T	N	B	F	K
5	I	H	Z	D	J	W	L	M	U	Y	K	L	G	K
6	O	R	N	A	U	P	T	D	V	Z	U	N	U	I
7	O	T	W	X	T	K	J	?	Q	E	O	S	O	K

The decoding step on one-time pad is simply converting the K4 text into numerical values from A=0 to Z=25 and subtracting the public key values from A=0 and Z=25 then doing modular arithmetic, take MOD26 so that any negative number will wrap around from A to Z (decoding is shifting back in reverse direction and encoding is shifting forward) for each position. This will now become the K4 public key for OTP to decode the 97 characters. The question mark is ignored. In the original order as K4, the result is:

	2	1	6	8	4	4	3	1	2	0	9	3	7	5
	1	2	3	4	5	6	7	8	9	0	1	1	1	1
1											X	S	D	V
2	S	E	X	W	W	B	P	L	A	Y	W	W	O	R
3	K	T	L	Q	Q	G	C	L	Y	Y	J	Q	F	O
4	R	M	Y	D	D	X	Q	M	K	L	T	I	M	J
5	J	V	E	X	R	Q	X	K	P	T	A	T	X	F
6	E	J	Q	B	E	R	L	I	N	C	L	O	C	K
7	C	R	K	H	E	Y	Z	W	I	V	U	N	O	X
8	O	K	R	E	A	W	K	M	H					

The plaintext will now read:

X SDVSE
 X WW B PLAY WW OR KTL QQ GCL YY J Q FOR MY DD
 X Q MK L TIM JJ VE
 X RQ
 X KP TAT
 X FE J Q BERLINCLOCK CR KHEYZ WIV U NO
 X OK REAWK MH

Now that the plaintext is shown, we can clearly see that it has been masked through various abbreviations and

omissions of letters to defend against computational brute forcing.

DISCUSSION & ANALYSIS

The entire approach to K4 was to determine a method in which Jim Sanborn, a sculptor, would be able to create an extremely challenging cipher without a significant amount of technical cryptography knowledge. Our goal was to reverse engineer a method where physical manipulation was possible in two dimensions and also three (i.e. Jefferson disk K4 public key).

While the techniques are not excessively sophisticated and unknown, the steps to decoding K4 is extremely laborious. This would not be possible without the aid of Microsoft Excel in terms of repeatedly manipulating, recalling, and decoding K4 through trial and error. Unmasking the K4 plaintext from one-time pad required some educated guesswork. We surmise that it is a radio transmission from a field operative and the corresponding response from headquarters. Our best translation of the masked text is attached in the figures at the end.

Possible K4 Third Hint & Multiple Plaintext Outcomes

The decoded plaintext has very few words actually available to be used as hints. Following Mr. Sanborn's patterns, he could choose from "SEX", "PLAY", or "WORK" or maybe pairs of letters. We may never know what his next hint would be. It may be also possible that using OTP could produce multiple outcomes of the plaintext. Our best attempts have resulted in one possibility, had Mr. Sanborn changed the hints, the outcome may have been different.

Possible Public Keys for K0-K3

Looking back on K0 through K3, it may be possible that all of the sections had some sort of public key available to decode the message. K0 used International Morse, K1 and K2 uses the Quagmire III Vigenere table on the other half of the Kryptos panel, while K3 may have some distinct markings to give clues that the section is actually a double columnar transposition versus a substitution. If this was the case, then a K4 public key pattern could have very well been inferred. We are unsure if Mr. Sanborn left clues to create a Jefferson-style disk to reconfigure K4 as a public key, but it is possible.

The "Berlin Clock" & K5 Segway

While there are many public clocks in Berlin, one of the clocks that would best fit the description would be the Alexanderplatz (Urania World Clock) from the K4 plaintext. At the base of the clock lies a compass rose which also matches the compass lodestone on the CIA grounds. It may be possible that the magnetic heading adjustment may require the physical presence along with a compass for K5.

CONCLUSION

Kryptos remains as an analog timepiece in the constantly evolving digital world. We believe the original intention of the artist was to utilize analog methods to experience a fraction of the joy of cryptography (and art) from Mr. Sanborn and Mr. Scheidt. The K4 section was intentionally future-proofed against digital (computer) brute force approaches and required the analog (human) touch. In our opinion, Kryptos remains a masterpiece for its physical manifestation of cryptography in American history. There is a progression of cryptographic methods coincides with key historical American eras. The intrinsic energy it took to physically and mentally create this sculpture cannot be even measured. Jim Sanborn has artistically chosen an elegant, yet simple way of encoding K4 while having its key hidden in plain sight.

While we believe our best interpretation of the decoded K4 section is a radio conversation between a field operative and headquarters, we may never know if there is an alternative darker possibility. Regardless of the outcome, it would be the true intent of an artist to leave some sort of esoteric meanings open to interpretation. With unlimited mediums available to an artist, art itself is the greatest form of cryptography.

FIGURES ATTACHED

Figure 1 – K4 Plain Text Phonetic Masking

Figure 2 – Outline of C² K4 Double-Mask-OTP

FIGURE 1:

K4 PLAIN TEXT PHONETIC MASKING

PT: X SDVSE X WW BPLAY WW OR CTL **QQ** GCL YY J **Q**
FORMYDD X **Q** MKLTIM JJ VE X RQ X

POSSIBLE MEANING:

*ADVISE. "B(RAVO)PLAY" OR K(EY)T(ASK)L(IST)??
G(ROUND)C(ONTROL)L(ANDING) J(ULIET)? FOR MY
D(EAD) D(ROP)? M(AR)K L(IMA) TIM(E),
V(ERNAL)E(QUINOX). RQ(SOS).*

PT: KPTAT X FE J **Q** BERLIN CLOCK CRK HEYZ WIV UNO
X OK REAWK MH

POSSIBLE MEANING:

*COPY T(H)AT. F(IELD) E(XPEDIENT) J(ULIET)?
BERLIN CLOCK CR(AC)K KEYS WITH (WHAT) YOU KNOW. OK
REWORK M(AGNETIC) H(EADING).*

FIGURE 2 - Outline of C² K4 Double-Mask-OTP (July 22, 2019)

